

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

Date	17 July 2026
Team ID	
Project Name	SB Stocks — MERN Stack Stock Trading Simulation Platform
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

The product backlog below breaks SB Stocks down into epics, user stories, and story points across four sprints:

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my username, email, and password.	2	High	
Sprint-1		USN-2	As a user, I receive a confirmation once registration is successful and am redirected to the login page.	1	High	
Sprint-1	Login	USN-3	As a user, I can log into the application by entering my registered email and password.	2	High	
Sprint-1		USN-4	As a user, I receive a JWT-based session on login so I stay authenticated across requests without re-entering credentials.	2	High	
Sprint-1	Dashboard	USN-5	As a user, I can view a dashboard summarizing my total net worth, available cash, and holdings value.	3	High	
Sprint-1		USN-6	As a user, I can view a live, auto-refreshing list of stocks with current price and % change.	5	High	
Sprint-1		USN-7	As a user, I can search the stock list by symbol or company name.	2	Medium	
Sprint-1		USN-8	As a user, I can view Top Gainers and Top Losers widgets on the dashboard.	3	Medium	
Sprint-2	Stock Detail	USN-9	As a user, I can open a stock's detail page to view its current price and live-feed status.	2	High	
Sprint-2		USN-10	As a user, I can switch between Candlestick, Trend, and Volume/Spread chart views across 1D/1W/1M/1Y ranges.	5	High	
Sprint-2	Trade Execution	USN-11	As a user, I can place a Buy order by selecting product type and entering a quantity.	5	High	
Sprint-2		USN-12	As a user, I can place a Sell order for shares I currently hold.	3	High	
Sprint-2		USN-13	As a user, I am shown the maximum number of shares I can buy based on my available cash.	2	Medium	
Sprint-2		USN-14	As a user, my Buy order is blocked if the requested quantity exceeds what I can afford.	3	High	

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3	Portfolio	USN-15	As a user, I can view my holdings with shares held, average purchase price, current price, and market value.	3	High	
Sprint-3		USN-16	As a user, I can see my total profit/loss and percentage return for each holding.	3	High	
Sprint-3		USN-17	As a user, I can view a portfolio-weighting donut chart broken down by stock symbol.	2	Medium	
Sprint-3	History	USN-18	As a user, I can view an order book listing every executed trade with status and timestamp.	3	High	
Sprint-3		USN-19	As a user, I can view a wallet ledger of my deposits and withdrawals.	3	Medium	
Sprint-3		USN-20	As a user, I can refresh my trade and audit logs on demand.	2	Low	
Sprint-3		USN-21	As a user, I can switch between Order Book and Wallet Ledger tabs on the History page.	2	Low	
Sprint-3		USN-22	As a user, I can view summary cards for total executed orders, buy capital invested, and sell capital recovered.	2	Medium	
Sprint-4	Profile & Wallet	USN-23	As a user, I can view my account profile — username, email, and authorization role.	2	Medium	
Sprint-4		USN-24	As a user, I can simulate depositing virtual funds into my wallet.	3	High	
Sprint-4		USN-25	As a user, I can simulate withdrawing virtual funds from my wallet.	3	Medium	
Sprint-4		USN-26	As a user, I can select a funding source (e.g., Bank Transfer) for a simulated deposit.	2	Low	
Sprint-4	UI/UX	USN-27	As a user, I can toggle Dark Mode for a more comfortable viewing experience.	2	Low	
Sprint-4		USN-28	As a user, I want the layout to be responsive across the dashboard, portfolio, history, and profile pages.	3	Medium	
Sprint-4		USN-29	As a user, I want clear form validation and error messages on trade and authentication forms.	3	High	
Sprint-4	Login	USN-30	As a user, I want my session to expire securely and require re-login after logging out.	2	Medium	

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date (Planned)	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	27 May 2026	01 Jun 2026	20	01 Jun 2026
Sprint-2	20	6 Days	03 Jun 2026	08 Jun 2026	20	08 Jun 2026
Sprint-3	20	6 Days	10 Jun 2026	15 Jun 2026	19	16 Jun 2026
Sprint-4	20	6 Days	17 Jun 2026	22 Jun 2026	20	22 Jun 2026

Velocity

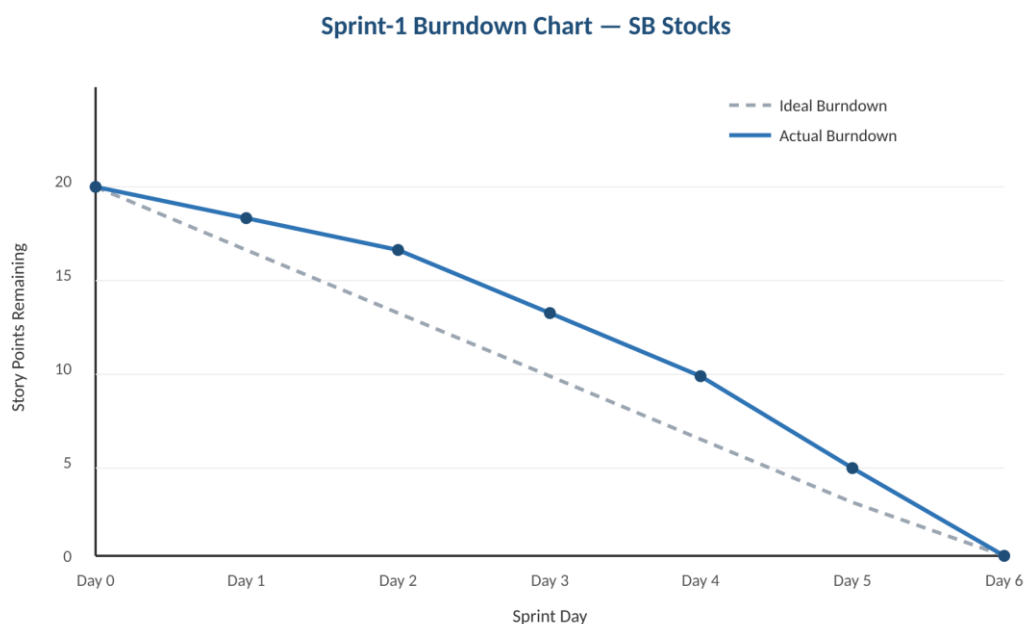
Across all four sprints, the SB Stocks team completed 79 of 80 planned story points, for an average velocity (AV) of:

$$AV = \text{Total Story Points Completed} \div \text{Number of Sprints} = 79 \div 4 \approx 19.75 \text{ points / sprint}$$

With a 6-day sprint duration, this is roughly $19.75 \div 6 \approx 3.29$ story points completed per day — a steady, sustainable pace that kept each sprint close to its 20-point plan.

Burndown Chart

A burndown chart is a graphical representation of work remaining versus time. It is widely used in agile methodologies such as Scrum, and can be applied to any project with measurable progress over time. The chart below tracks Sprint-1 of SB Stocks, where the team registration, login, and dashboard stories were completed essentially on the ideal pace, finishing all 20 points by Day 6:



Sprint-1 burndown: ideal vs. actual story points remaining, Day 0 (sprint start) through Day 6 (sprint end).

Reference: <https://www.atlassian.com/agile/tutorials/burndown-charts>

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>